Diesel Engine
V 396 TC 64
V 396 TE 64

Engine for Marine Propulsion

Maintenance Schedule
M050384/20E

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Low operating and maintenance costs as well as operational availability and reliability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.
The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble–free engine operation at all times. For this purpose, always:

- ensure that sufficient fuel is available
- ensure that the combustion air is dry and clean
- use only dry, clean compressed air
- use only clean, filtered raw water

Moreover, it is essential that

- maintenance services be carried out by trained personnel
- suitable tools be used
- genuine spare parts and fluids and lubricants as per the current MTU Fluids and Lubricants Specification A001061 be used.

Our Product Support Service will always be available should assistance be required.
Preventive maintenance instructions

- Special care should be taken to keep the power plant in a clean and serviceable condition at all times to facilitate detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic parts from oil and fuel, never treat with organic detergents. Wipe with dry cloth only.

MTU Maintenance Concept
The MTU maintenance concept is a preventive maintenance concept and features the W1 to W6 Maintenance Echelons W1 to W6 as outlined below.
Preventive maintenance permits advance operational planning and increases availability.

Main Features of Maintenance Tasks for MTU Engines

Maintenance Echelon	W1:	Daily operational checks.
Maintenance Echelons	W2, W3 and W4:	Periodic maintenance services to be performed during out-of-service periods without the need for engine disassembly.
Maintenance Echelon	W5:	This echelon requires partial engine disassembly.
Maintenance Echelon	W6:	Major overhaul. This echelon requires complete engine disassembly.

The time intervals according to which W2 to W6 maintenance operations are to be performed and the relevant inspection and maintenance tasks are based on operational experience. They have been determined so as to ensure correct engine operation until the next maintenance work.
Particular operating conditions may require the Maintenance Schedule to be altered to compensate.

NOTE
The five-digit codes on the following pages indicate the groups and subgroups in the Engine Operation Manual (Part G - Task Description) in which the tasks involved are described.
This Maintenance Schedule may include components which are not installed on your engine; these may be disregarded.

Application Group

1A: Vessels with unlimited operating range and/or unrestricted continuous operation

Standard Load Profile

Load referenced to fuel stop power	%:	100	90	< 15
Time element	%:	10	80	10

Maintenance Frequency Chart

Maintenance	W1:	Maintenance echolon	daily
	W2:	Operating hours	250
		Limit (months)	6
	W3:	Operating hours	1 000
		Limit (years)	1
	W4:	Operating hours	2 000
		Limit (years)	2
	W5:	Operating hours	12 000
		Limit (years)	6
	W6:	Operating hours	24 000
		Limit (years)	18

Maintenance Schedules

Code	One-time additional services after first 50 operating hours - on a new engine, or after W5 or W6 maintenance	
G06.101	Valve gear	Check valve clearances, adjust if necessary
G10.111	Intake air system	Check tightness of securing bolts
G10.531	Exhaust system	Check tightness of securing bolts Check for leaks
G12.311	Fuel prefilter	Clean
G12.311	Fuel prefilter	Replace filter elements
G12.321	Fuel duplex filter	Replace filter elements
G13.112	Engine coolant pump	Check relief bore for obstructions
G13.123	Raw water pump	Check relief bore for obstructions
G13.511	Bilge pump	Check relief bore for obstructions
G14.019	Engine coolant	Flush strainers in coolant return lines
G16.002	Engine oil	Take sample and analyze
G16.121	Engine oil edge-type filter	Operate handle
G19.011	Engine mounting	Check tightness of securing bolts
G19.011	Gearbox mounting	Check tightness of securing bolts
G88.101	PTO systems, main/aux. PTO ends	Check tightness of securing bolts

Code	Maintenance Echelon W1 - Monitoring	
G10.051	Exhaust system	Check exhaust gas colour Drain condensate - if drain valve provided
G10.181	Intercooler	Check drain line for water discharge and obstructions
G10.211	Air filter	Check contamination indicator
G10.401	TC control	Drain condensate from actuating cylinder
G12.001	Fuel	Check supply
G12.311	Fuel prefilter	Operate handle
G12.311	Fuel prefilter	Drain water and contaminants
G14.011	Engine coolant	Check level
G16.002	Engine oil	Check level
G84.001	Monitoring system	Carry out lamp test
G84.002	Engine operation	Check running noises Check engine and external pipework for leaks Check engine revolutions, pressures and temperatures - if gauges installed
G86.051	Compressed air	Check operational pressure Drain condensate
G88.831	Pedestal bearing	Check oil level